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# AN EXPERIMENTAL OBSERVATION OF A SOLITARY WAVE IMPINGEMENT, RUN-UP AND OVERTOPPING ON A SEAWALL*

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**Abstract:** A sequence of laboratory experiments using solitary waves was performed to model the effect of leading form of three types of tsunamis (a bore, an impinging wave and an overtopping wave) on a seawall on a sloping beach. The wave evolution process, impinging pressure along the seawall surface, total overtopping discharge behind the seawall and the maximum run-up height on the rear slope were measured and compared. Laboratory data were employed to re-examine relevant empirical formulae in the literature. The effect of the presence of the seawall in reducing maximum run-up height using the present setup was briefly discussed. The present data can be used for calibrating numerical and mathematical models.

**Key words:** solitary wave, tsunami, wave force, run-up, overtopping

## Introduction

The leading form of a tsunami-wave train in the shallow water region forms either a sequence of bores or plunging type breaking waves that strike a nearshore region[1]. Furthermore, a very large tsunami may overtop and sweep onshore infrastructure on rare occasions[2]. Figure 1 gives a field evidence of the damage due to the Indian Ocean tsunami on Boxing day in 2004[3], where it can be clearly observed that a combination of run-up inundation and debris flow resulting in large wave forces and local scours that significantly damaged nearshore infrastructures. The impinging wave force and overtopping damage subject to a tsunami wave are of particular importance for planning a tsunami evacuation policy. In particular, the previous studies have been shown that the most popular gravity water-waves employed in studying tsunami behavior are either N-wave or solitary wave since they physically exhibit an analogous wave propagation behavior of the leading form of tsunami waves[4]. Tadepalli and Synolakis[5] mathematically justified that a leading-depression N-wave climb

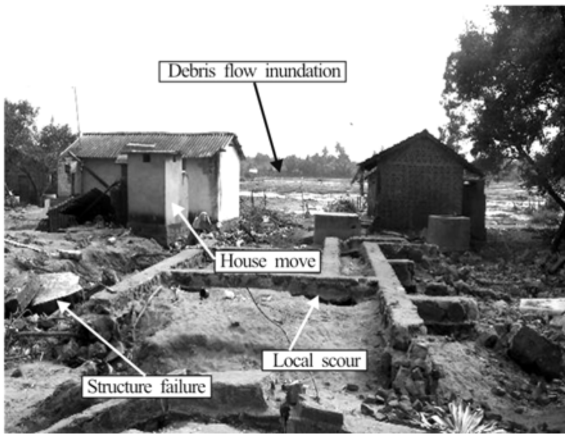


Fig.1 Field survey evidences in Sri-Lanka after the Boxing Day tsunamis in 2004 (http://coastal.usc.edu/plynett/Sri%20Lanka%20Tsunami%20Damage/slides/d2_house_removed.html)

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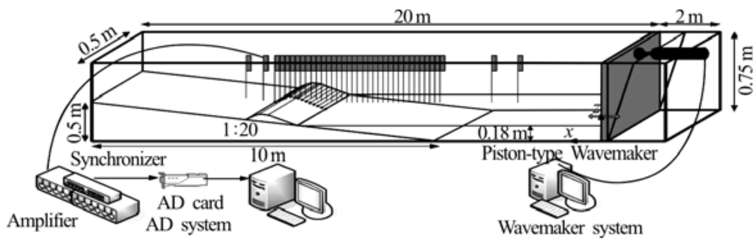


Fig.2(a) Schematic diagram of the laboratory flume and the corresponding measurement facility

**Table 1 Experimental conditions**

| Tsunami type | Experimental trial | Symbol | $h_o$ (m) | $\varepsilon = H_o / h_o$ |
|---|---|---|---|---|
| | 1-59 | A | 0.16 | 0.22, 0.25, 0.29, 0.32, 0.35 |
| Bore | 60-119 | B | 0.18 | 0.22, 0.25, 0.29, 0.32, 0.35 |
| | 120-179 | C | 0.2 | 0.22, 0.25, 0.29, 0.32, 0.35 |
| Impinging wave | 180-206 | D | 0.22 | 0.1, 0.22, 0.29 |
| Overtopping wave | 207-224 | E | 0.256 | 0.1, 0.23 |

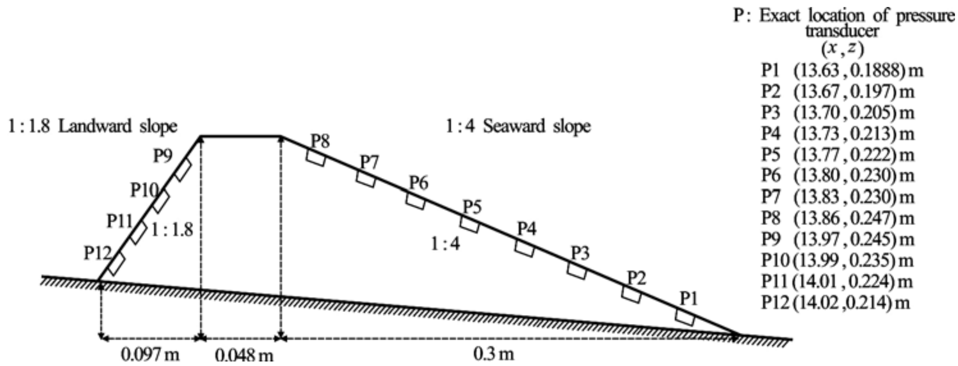


Fig.2(b) Schematic diagram of seawall model and the exact locations of each pressure transducer

higher on a sloping beach than a leading-elevation N-wave, which is able to be applied to the research of nearshore-generated tsunamis. Carrier et al.[6] further derived an asymptotic theory for describing run-up and draw-down subject to different types of leading tsunami forms (see Synolakis and Bernard[4] and Hsiao et al.[7] for more review). However, an integrated study concerning a solitary wave interact against a seawall is relatively limited[8].

The main aim of this paper is to present a set of new laboratory experiments using solitary waves to model impingement, overtopping and run-up of three types of tsunamis (a bore, an impinging wave and an overtopping wave) interacting with a seawall upon a sloping beach. We shall report systematical laboratory data for observing the wave evolution process and measuring wave pressure loading due to different waves interacting against a seawall, to which the wave overtopping discharge and subsequent inundation height on the back slope (i.e., maximum run-up height) are quantitatively measured. The resulting laboratory data are employed to re-examine relevant empirical formulae in the literature.

## 1. Experiment

Laboratory experiments were carried out in a two-dimensional wave flume with 22 m long, 0.5 m wide and 0.75 m deep at the Tainan Hydraulics Laboratory (THL) of National Cheng Kung University (Fig.2(a)). The glass sidewalls were equipped on both side of the flume to facilitate visual observation and video recording. Target solitary waves with negligible tail oscillation were generated using an algorithm developed by Goring[9] to model the three-type leading waves of a tsunami (see Table 1) via a programmable piston wave-paddle. The analysis system is based on a Cartesian coordinate system, where $x$

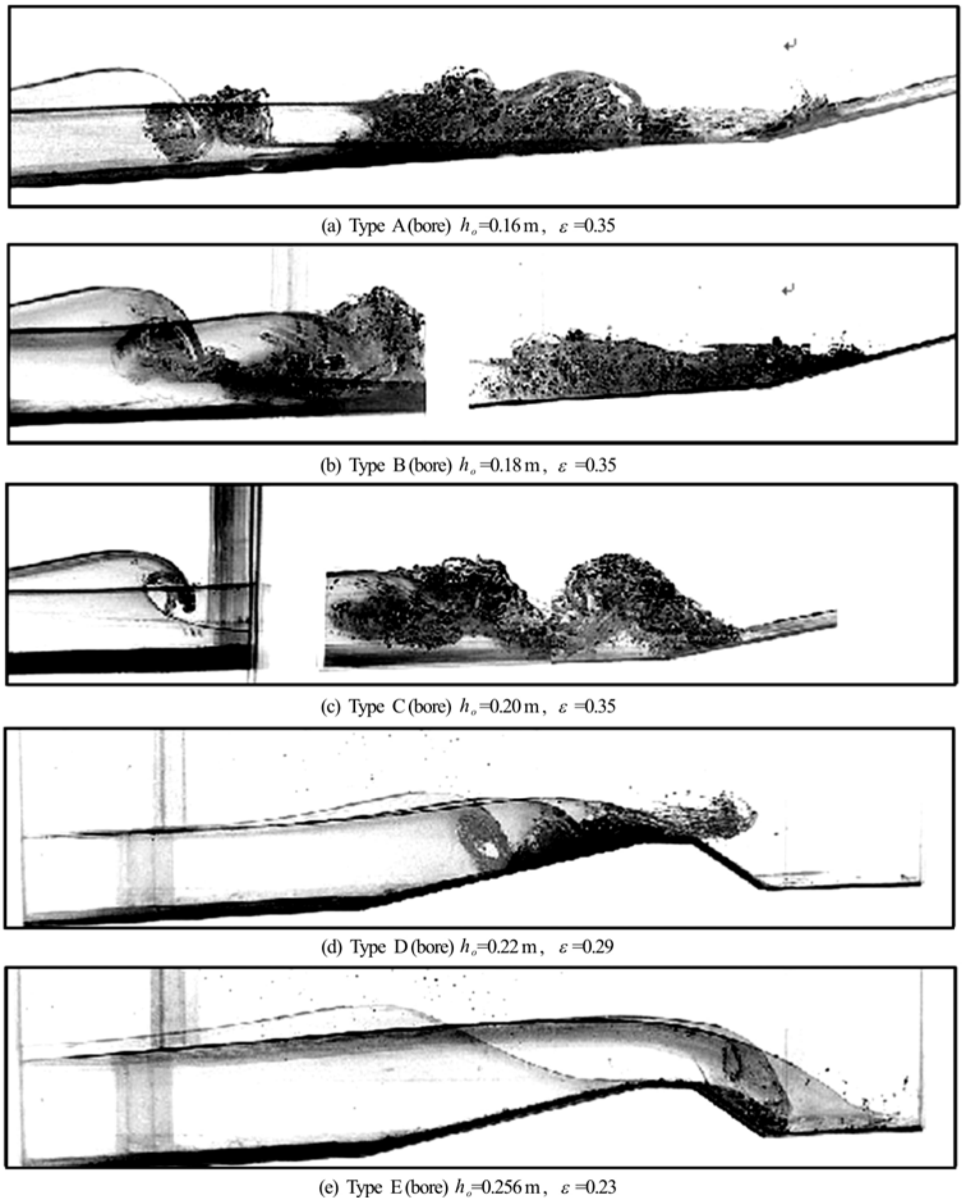


Fig.3 Laboratory sequences of different types of tsunamis impinging and overtopping the seawall. The wave propagates from left to right

and $z$ are defined as the abscissa and ordinate with positive directions downstream the wave-paddle and upward from the flume bottom, respectively.

The experimental bathymetry and topography was composed of two parts (see Fig.2(b)). One is a 1:20 sloping beach made of Plexiglas starting at 10 m from the wave-paddle (i.e., $x = 10$ m), the other is a seawall modeled by a trapezoidal impermeable caisson built of aluminum with a front 1:4 and landward 1:1.8 slopes starting at a distance of 3.6 m from the beach toe (i.e., $x = 13.6$ m). Note that Plexiglas material of the sloping beach and the seawall model was selected for reducing the effect of bottom friction. The borders between the seawall model, the sloping beach and the glass wall were filled with silicone to avoid fluid infiltration. The local free surface deformation $\eta$ was tracked via wave gauges deployed along the flume (see Fig.2(a)), the wave impinging pressure $P$ was measured using the pressure transducers (0.003 m/ each, ST-type by Japan) exactly burying upon the seawall surface (see Fig.2(b) for locations). The data sampling frequencies of free surfaces and pressure loads were selected by 50 Hz and 100 Hz, respectively. Note that the accuracy of these sampling rates was validated by a set of preliminary tests using higher sampling rates, no noticeable difference in measurement data was found. The total overtopping discharge per unit width $q$ was obtained via the

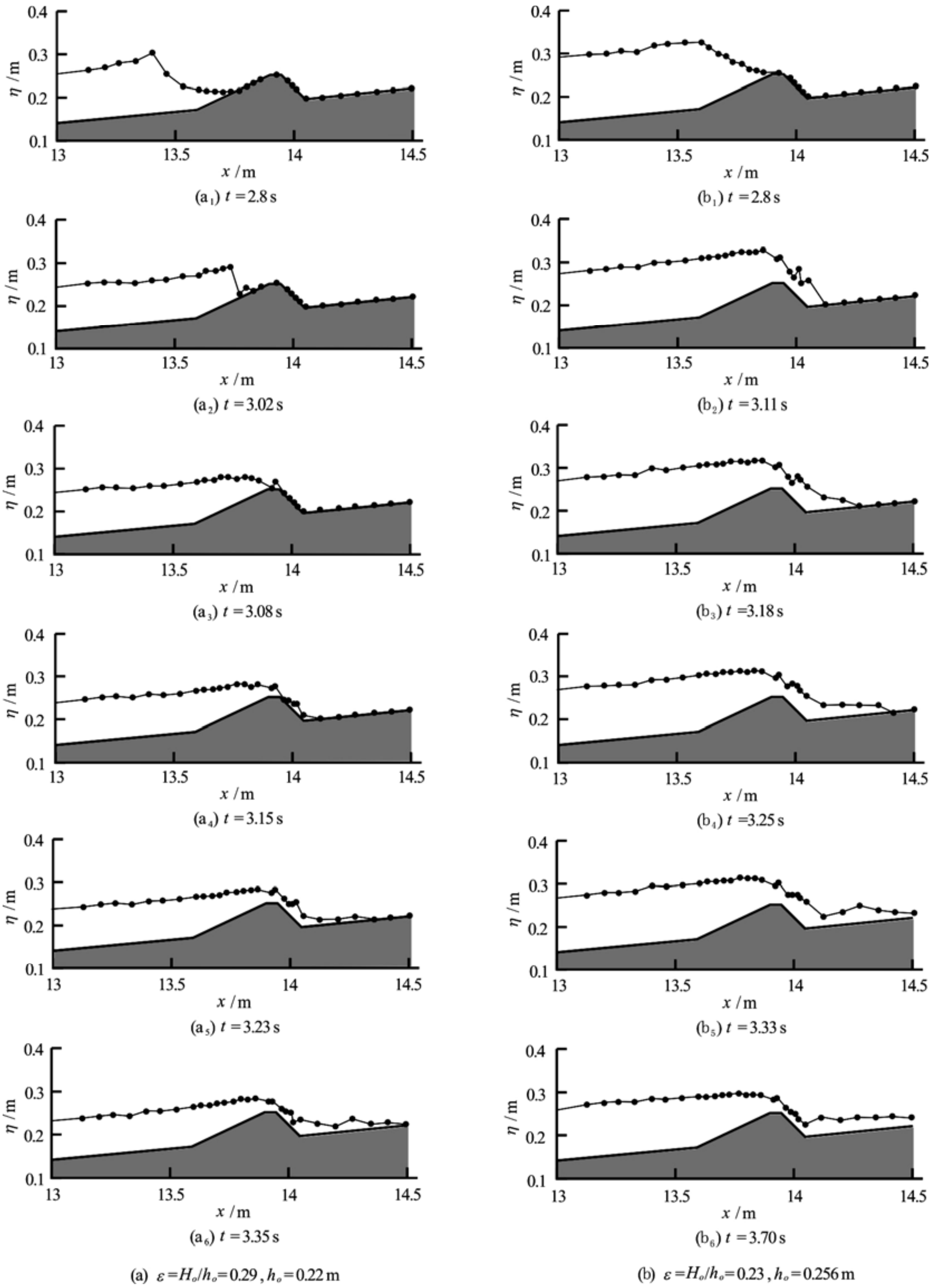


Fig.4 Spatial snapshots of Types D (left column) and E (right column) tsunamis impinging and overtopping the seawall. The free surface evolution is plotted at dimensional time

local water depth $h$ at a steady state recorded by gauges behind the seawall using a trigonometric expression associated with the present experimental topography (i.e., $q = 21.8h^2$ ). The maximum run-up height $R$ was defined as a vertical distance from the still water level to the position on the landward slope where the overtopping wave reached.

Table 1 categorizes 224 laboratory trials of the three-type tsunami behaviors (i.e., Types A, B, C: bore, Type D: impinging wave, and Type E: over-

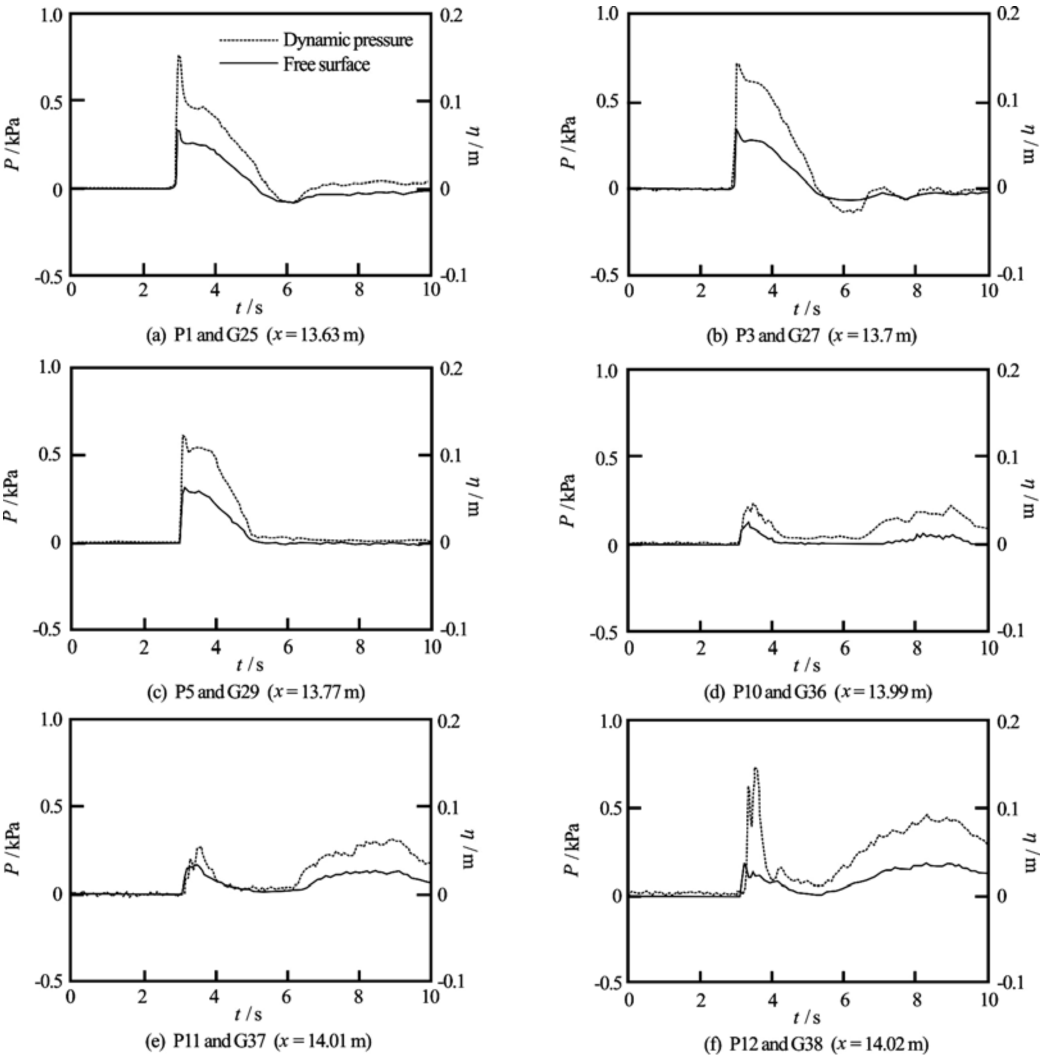


Fig.5 Variation of dynamic pressure and the corresponding free surface elevation at the same positions upon seawall surface (Type D: $h_o = 0.22$ m , $\varepsilon = 0.29$ )

topping wave) with varying offshore water depths, $h_o$, and incident wave nonlinearity, $\varepsilon = H_o / h_o$, in which $H_o$ denotes offshore wave height. Note that in all experimental trials, a reference wave gauge was fixed at $x = 8.9$ m. However, it should be noted that the wave evolution processes of some cases with thin overtopping flows were significantly altered by crowed wave gauges locating at certain locations, i.e., Types A and B: $h_o = 0.16$ m and 0.18 m. Three identical tests were therefore carried out for each wave condition (those tests were also included in the 224 laboratory trials listed in Table 1) to measure overtopping discharge and run-up behaviors by placing wave gauges of a minimum requirement. A maximum deviation of 3% in the reference wave height $H_o$ was satisfactorily given for the same wave conditions in different experimental runs.

## 2. Results and discussion

### 2.1 *Tsunami wave type*

Figure 3 depicts a sequence of overlapped laboratory images taken by a high-speed camera (Nikon COOLPIX5700) with a frame acquisition rate of 3 Hz for three-type tsunamis in terms of (1) wave breaking and bore formation in front of the seawall (Types A, B and C), (2) wave breaking and impingement at the front face of the seawall (Type D), and (3) wave overtopping behind the seawall (Type E). With regard to Types A, B and C, the waves breaking offshore near the shoreline and subsequently form a sequence of turbulent bores to impinge and overtop the seawall. Comparing to Types D and E, the waves straightforwardly collapse and overturn upon the seawall crown,

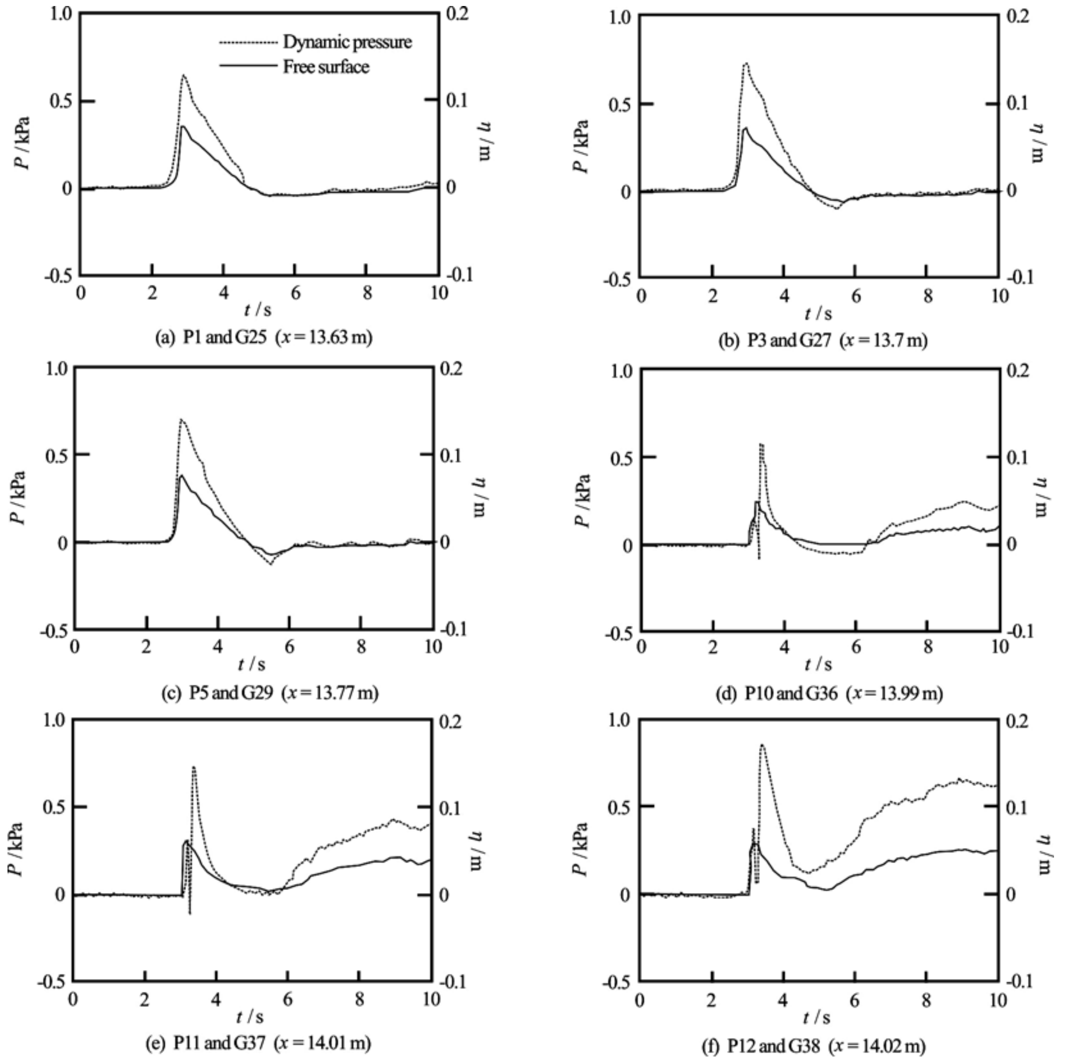


Fig.6 Variation of dynamic pressure and the corresponding free surface elevation at the same positions upon seawall surface (Type E: $h_o = 0.256$ m , $\varepsilon = 0.23$ ). Note that the location of sub-atmosphere pressure is pointed out by a black arrow

respectively. Note that capitals (i.e., A-E) will be used to represent the cases mentioned above in order to simplify the following descriptions.

Apparently, the cases of A-C break as a plunging breaker, in which the wave fronts curl over and also capture considerable air into front quiescent fluid. The breaking waves of A and B initially burst into the front still water and subsequently form so-called "splash-up" behavior[10] as turbulent bores to propagate onshore. Interestingly, the breaking wave of C exhibits a different behavior compared to A and B, in which a considerable "wedge-shape" bubbly fluid block is formed to propagate inland. In the laboratory, we have observed that the overtopping flow of A was relatively weak comparing to that of B and C (not shown in the pictures). The overtopping flows of B and C were to a spray jet and the overtopping flow of C is more violent than that of B. In particular, even though the entrapped air in bores is obvious and might contribute to some experimental uncertainties in measurements[11], the maximum pressure distributions along the seawall surface are quite similar and the corresponding comparisons will be shown herein.

Regarding D, the seawall contributes to strong reflection to make re-collisions of coming breaking waves and reflected jets dramatic, a stronger overtopping flow is also observed to spill on the inland onshore slope. The wave straightforwardly collapses behind the breakwater in E. The laboratory observation indicates that a strong vortex would be generated near the toe of leeside seawall, which initially climbs toward the crown and gradually propagates downstream because of fluid advection. Interestingly, our experimental data point out that the presence of vortex would capture significant air-bubbles, which occasionally contributes to the reduction of wave pressure loadings (not shown in this paper).

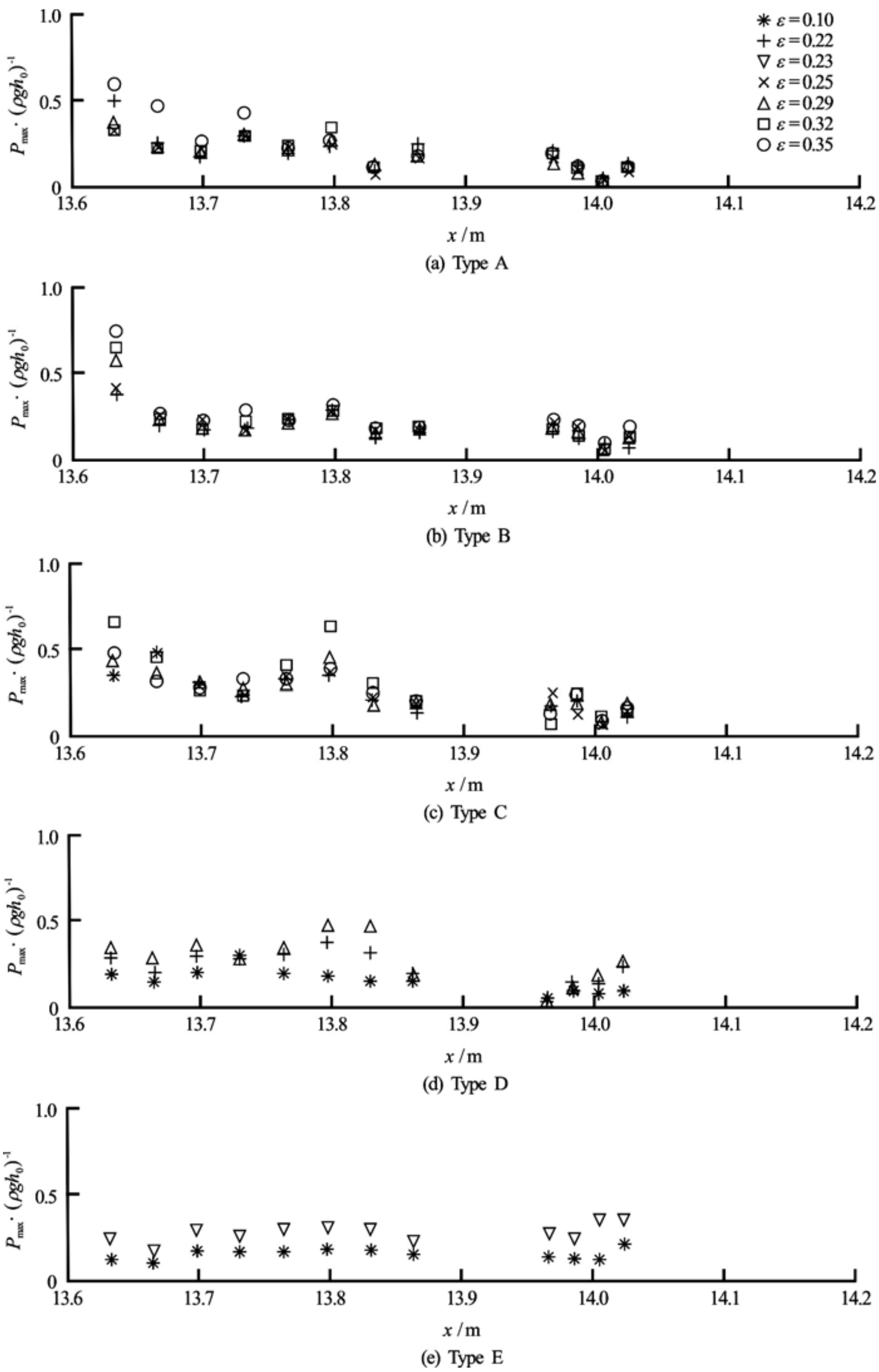


Fig.7 Comparison of maximum wave impinging pressure distributions along the seawall surface

2.2 *Free surface and wave impinging pressure*

Figure 4 presents spatial snapshots of two representative cases: Type D and Type E. It is observed that both overtopping waves collapse behind the seawall and the surface displacements are obvious (i.e., Figs.4($a_4$)-4($a_6$), Figs.4($b_2$)-4($b_5$)], in which strong vortexes appear and also capture considerable air into the flow fields. Both of them initially climb toward the crown and gradually spread downstream due to advection. Note that the experimental observation reveals that the reverse flow of E would form a second breaking wave like a hydraulic jump to strongly interact with the coming tail waves.

Figures 5 and 6 present the comparisons of D and E on the local free surface and wave impinging pressure measuring at the same position along the seawall surface. It can be seen that the time history of wave impinging pressure has an inclination similar to that of free surface motion, in which a sudden spike and a gradual decay under wave impinging are also found. Particularly, a so-called "sub-atmosphere pressure" behavior (pointed out by a black arrow in Fig.6) reported by Bullock et al.[12] is clearly observed in our results (see Figs.5(e), 5(d) and 5(f), 6(e) and 6(f)). Furthermore, the pressure loading has been persisted in acting upon the seawall, which has been generated in a high aeration fluid region. In this region severe damping oscillation following the pressure spike is interestingly found to decrease down to negative (e.g., Figs.6(d) and (f) at $2\text{ s} < t < 4\text{ s}$).

Figure 7 summarizes the maximum wave impinging pressure distributions along the seawall surface for all experimental trials. Note that the pressure data are normalized by the corresponding hydrostatic pressures (i.e., $\rho g h_o$, where $\rho$ is the density of freshwater and $g$ is the gravitational acceleration) for a comparison. It is found that the maximum impinging pressures of A, B and C always occur near the toe of breakwater, where the magnitudes are almost of the same order of 60%-70% compared to the corresponding hydrostatic pressure. However, for C the maximum wave impinging pressure loading moves to an upper position of front slope ($x = 13.8\text{ m}$ in Fig.7(c)). For D and E the maximum pressure variations are relatively mild compared to those of A, B and C. It is also observed that behind the seawall the pressure loading increases almost linearly with the decrease of transducer elevation (see Fig.7(d)).

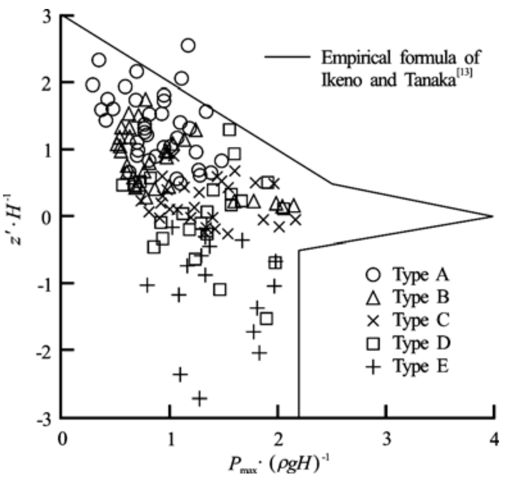


Fig.8 Vertical distribution of the maximum dynamic wave pressure along the seawall

Figure 8 further presents the vertical variation of maximum pressure load upon the seawall. The empirical formula (Eq.(1)) reported by Ikeno and Tanaka[13] is also plotted for comparison. Note that this formula suggests that the maximum pressure load occurs near the still water level, i.e., $z'/H' = 0$, where $z' = 0$ and $H'$ are defined as the position at the still water level and the local wave height recorded at the toe of seawall (i.e., $x = 13.6\text{ m}$), respectively. Interestingly, our data exhibit a similar inclination compared to Eq.(1), suggesting that the formula is quite reliable for estimating the vertical maximum pressure loading of the present seawall setup.

$$P_{\max} / \rho g H = 3 - \frac{z'}{H'}, \quad 0.5 \le \frac{z'}{H'} \le 3 \tag{1a}$$

$$P_{\max} / \rho g H = 4 - \frac{3z'}{H'}, \quad 0 \le \frac{z'}{H'} \le 0.5 \tag{1b}$$

$$P_{\max} / \rho g H = 4 + \frac{3.6z'}{H'}, \quad -0.5 \le \frac{z'}{H'} \le 0 \tag{1c}$$

$$P_{\max} / \rho g H = 2.2, \quad \frac{z'}{H'} \le -0.5 \tag{1d}$$

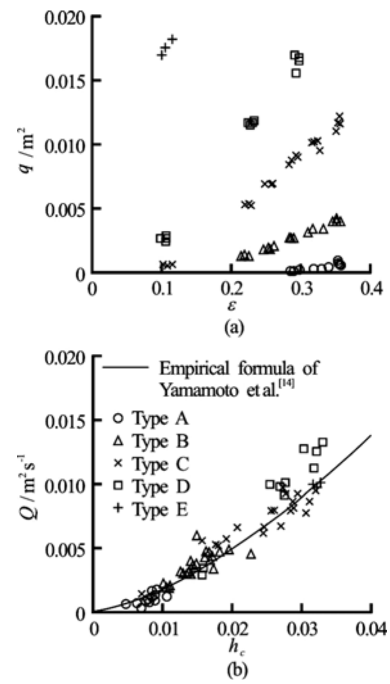


Fig.9 Laboratory data of total overtopping discharge per unit width

## 2.3 *Overtopping discharge and effect of presence of seawall on maximum run-up height*

The main goal of installing a seawall is to protect inland region from wave attacks. The maximum run-up height and overtopping discharge are two important indices for designing a breakwater. Figure 9 shows the measured overtopping discharge per unit width (Fig.8(a)) and the overtopping flow rate per unit width as well as unit time $Q$ (Fig.8(b)). The empirical formula (Eq.(2)) for evaluating a tsunami overflow rate suggested by Yamamoto et al.[14] is also plotted for a comparison, in which $h_c$ denotes the overflow height (i.e., incident wave height minuses the break-water height) and 0.55 is an empirical value suggested by Yamamoto et al.[14]. Our experimental data $q$ are obtained by computing the duration and the surface variation between two gauges placed on the seawall crown.

$$q = 0.55\sqrt{gh_c^3} \tag{2}$$

Overall, the measurement data suggest that the overtopping discharge monochromatically increases with the increase of wave nonlinearity. It should be note that the overtopping discharge will be a constant for the cases with $h_o = 0.256$ m and $\varepsilon > 0.22$ because of the present bathymetry. Equation (2) is believed to reasonably capture the overflow rates of present experiments.

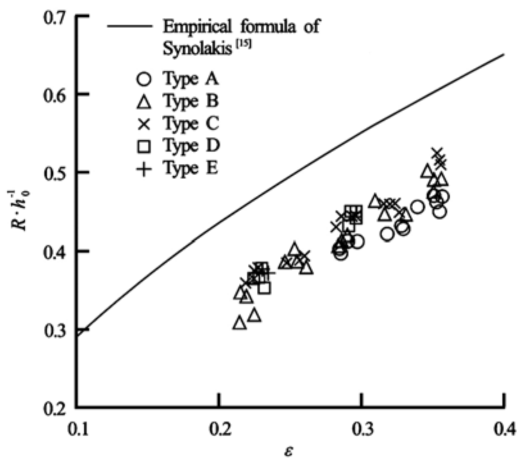


Fig.10 Comparison of maximum run-up height with/without the presence of seawall

Figure 10 compares the run-up data against the Well-known empirical formula (Eq.(3)) reported by Synolakis[15]. This formula was found via comprehensive laboratory data concerning the run-up heights subject to breaking solitary waves climbing up a 1:19.85 sloping beach. Our data show that the run-up heights can be significantly reduced by approximately 20% in case the present seawall is installed.

$$\frac{R}{h_o} = 1.109\left(\frac{H_o}{h_o}\right)^{0.582} \tag{3}$$

## 3. Concluding remarks

In this paper, a sequence of laboratory experiments is presented to model three type tsunamis (a bore, an impacting wave and an overtopping wave) impinge and overtop a seawall upon a sloping beach. The wave evolution courses, impacting pressure load, overtopping discharge and maximum run-up height are quantitatively measured and reported. Our analyses indicate that the maximum wave impinging pressure could reach up to approximately 60%-70% compared to an offshore hydrostatic pressure. The entrapped air in breaking waves would lead the "sub-atmosphere pressure" phenomenon to decrease the pressure loading. The data also exhibit that the maximum pressure being observed near the still water level and the empirical formula of Ikeno and Tanaka[13] is quite reliable. The overtopping discharge is found to monochromatically increase with the increase of wave nonlinearity, and is well predicted comparing to the formula of Yamamoto et al.[14]. The setup of present seawall is capable of reducing the maximum run-up height up to approximately 20% comparing to that without any seawall being installed. Importantly, the data presented in this paper could be employed to calibrate future mathematical and numerical models particularly on solitary wave/tsunami wave research. Further researches using an armor breakwater[16] as well as a flow visualization technique to investigate the topics on impacting wave force subject to air entrapment, location of wave impingement, corresponding flow characteristics are highly recommended.

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